

6 자유도 자기부상시스템의 고정밀 궤적 추종을 위한 오프셋-프리 모델 예측 제어 Offset-free Model Predictive Control of 6-DOF Magnetic Levitator for High-precision Trajectory Tracking

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In this work, we present an offset-free model predictive control (MPC) strategy to enhance the trajectory tracking performance of a 6-degree-of-freedom (DOF) magnetic levitation system. Classical motion control methods, such as PID controllers, lead-lag compensators, and linear quadratic regulators (LQR), are widely used due to their intuitive characteristics and ease of implementation. However, they often suffer from performance degradation under large-step reference inputs, primarily due to current saturation, which requires explicit consideration of input constraints. To address this limitation, we implement an MPC framework that explicitly handles input constraints while predicting future behavior of the system. The 6-DOF magnetic levitator is modeled as a second-order linear model with a lumped disturbance, and a disturbance observer is employed to estimate states and disturbances. The predicted states are then used to optimize the control inputs in real time. Computational delay is accounted for by predicting the system state during the computation interval while maintaining the previous control input. Experimental results show that the proposed strategy outperforms classical controllers, improving transient behavior of the system without steady state error, while respecting input constraints at 1 kHz sampling rate. These results validate that the proposed approach is suitable for mechatronic systems with fast dynamics.

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